Milestone 1: Data Loading and Cleaning:

Tasks:

1. Define objectives, KPIs, and understanding workflow

Project overview:

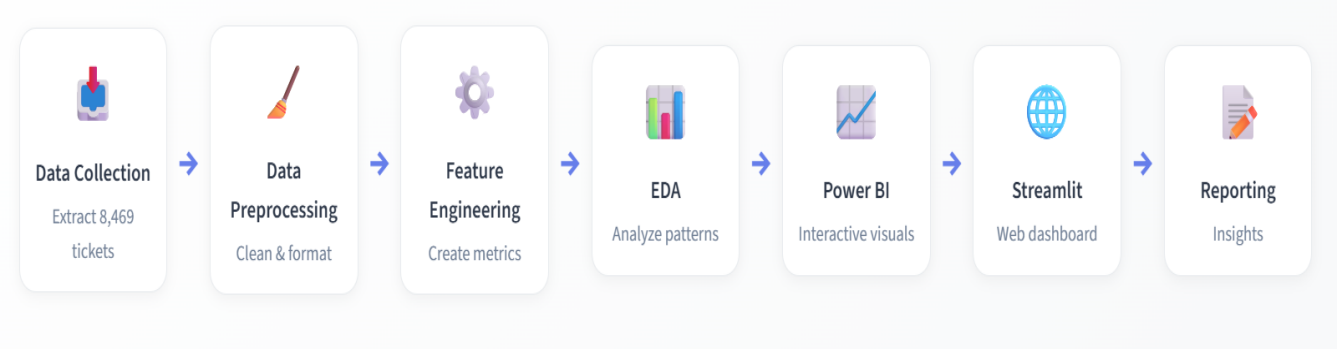
The objective of this project is to analyze IT support ticket data to identify key performance trends, optimize resolution times, and enhance service efficiency through data analysis and visualization. The goal is to uncover patterns in customer requests, technical issues, and support performance metrics to recommend improvements in workflow and resource allocation.

Business insights:

- Understand and preprocess IT support ticket data for analysis.
- Explore trends in ticket volume, priority, resolution time, and category distribution.
- Identify clusters of similar issues using similarity scores and clustering techniques.
- Visualize performance metrics through dashboards and charts.
- Derive actionable insights to optimize IT support operations.
- Summarize findings in a final analytical report and presentation

Project Workflow:

- **Data Collection** - Extract 8,469 tickets from internal IT support system
- **Preprocessing** - Clean, format, and validate dataset for analysis
- **Feature Engineering** - Create derived metrics like resolution categories and age groups
- **EDA** - Analyze patterns and identify trends using statistical methods
- **Power BI** - Build interactive visuals for business intelligence reporting
- **Streamlit** - Develop web dashboard with real-time filtering capabilities
- **Reporting** - Generate actionable insights for stakeholders



2. Load the dataset using pandas and Explore schema and data types

Dataset overview :

- Dataset Source from Internal IT Support Ticket System dataset.
- The dataset covers the time period from 2020 to 2021, containing two years of ticket data.
- Total Tickets 8,469 included in the dataset.
- Ticket attributes:Ticket ID,Type,Status,Priority,Channel.
- Customer information:Age,Gender,Country.
- Product information:Product Purchased.
- Performance metrics:First Response Time,Resolution Time,Satisfaction Rating.

3. Data Cleaning-missing values, incorrect data in text fields

4. Calculate initial ticket distribution by Type, Priority, and Category

5. Create new features: Resolution_Duration and Priority_Score

- Resolution_Duration = Time to Resolution- First Response Time
- Priority_Score = {"Low": 1,"Medium": 2,"High": 3,"Critical": 4}

Deliverables:

- Cleaned dataset
- Feature engineering summary
- Data dictionary